# Potential Energy and Kinetic Energy of a body executing SHM

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# Whatever we will discuss

- Mathematical Expression for-

  a) Displacement
  b) Velocity
  c) Maximum Velocity
  d) Acceleration
  e) Maximum Acceleration
  f) Time Period
  g) Frequency
  h) Angular Frequency

- Potential Energy of a body executing SHM
- Kinetic Energy of a body executing SHM
- Total Energy of a body executing SHM
- Related mathematical problems

# Mathematical Expression for-

a) Displacement $$y = A\sin(\omega t + \phi)$$

b) Velocity $$v = \frac{dy}{dt} = \omega\sqrt{(A^2 - y^2)}$$

c) Maximum Velocity $$v_{\max} = \omega A$$

d) Acceleration $$a = \frac{d}{dt}\left(\frac{dy}{dt}\right) = -\omega^2 y$$

e) Maximum Acceleration $$a_{\max} = -\omega^2 A$$

f) Angular Frequency $$\omega \left(= 2\pi f = \frac{2\pi}{T}\right) = \sqrt{\frac{k}{m}}$$

g) Time Period $$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

h) Frequency $$f = \frac{\omega}{2\pi} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

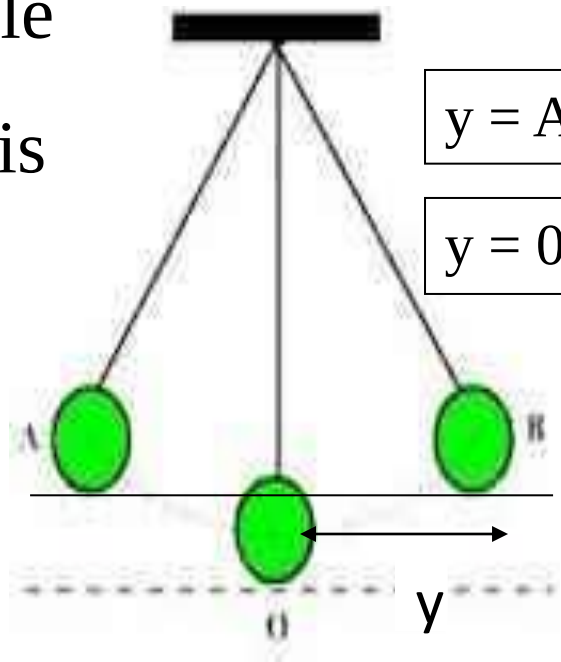
# Velocity of a body executing SHM

The displacement of a particle executing simple harmonic motion is

$$y = A\sin(\omega t + \phi)$$

$$\frac{dy}{dt} = v = \omega\sqrt{A^2 - y^2}$$

$$v_{\max} = \omega A$$




When $y = 0$, $v = max$ & When $y = A$, $v = 0$

## Problem:

- The displacement of moving particle at any time t is given by $y = a\cos\omega t + b\sin\omega t$ . Show that the motion is simple harmonic.
- A body is vibrating with simple harmonic motion of amplitude 15cm and frequency 4Hz. Calculate the maximum value of velocity and the velocity when displacement is 9cm. **Ans.: 376.99 $cms^{-1}$ and 301.59 $cms^{-1}$**

# Acceleration of a body executing SHM

Displacement equation

$$y = A\sin(wt + \phi)$$

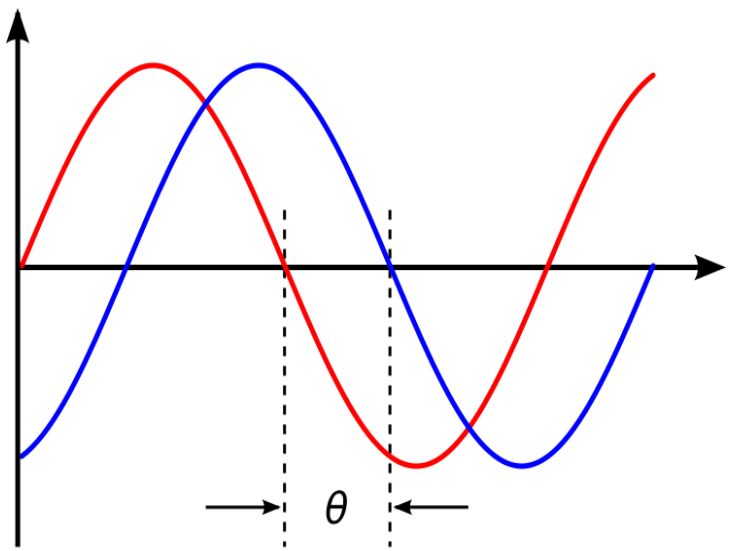
$$\frac{d^2y}{dt^2} = a = -\frac{k}{m}y$$

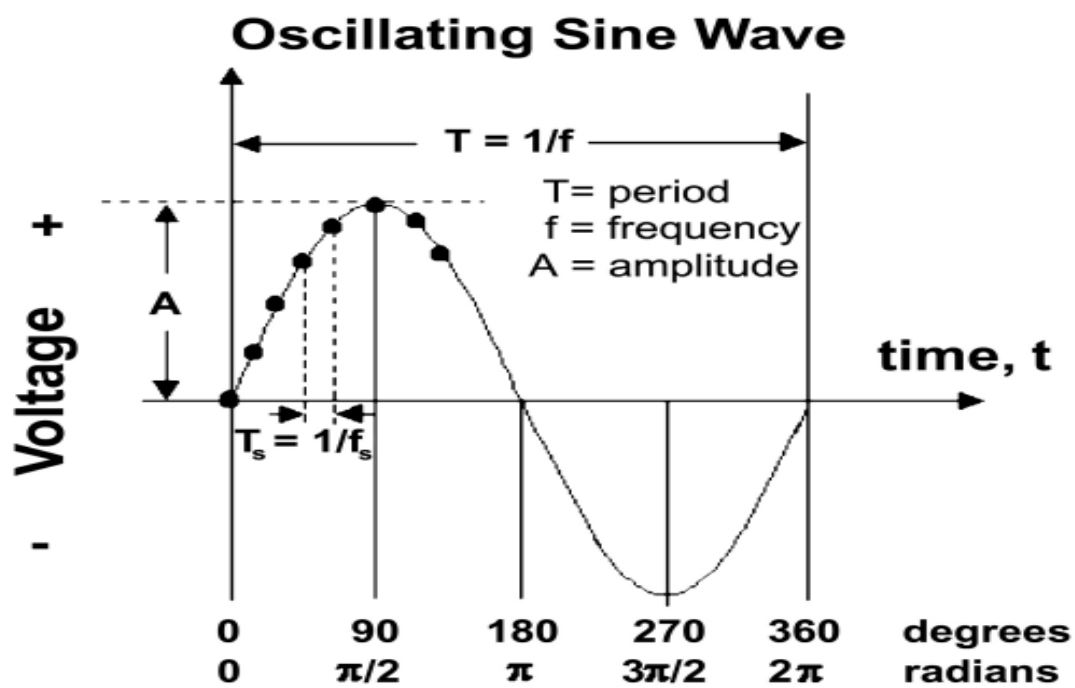
y = A, acceleration = max

The maximum value of acceleration occurs at the position of one of its extreme displacements ( y = A ).

# Phase of a simple harmonically vibrating particle

In the equation $y = A \sin(\omega t + \theta)$, the term $(\omega t + \theta)$ is known as the phase. At t=0, the term = $\theta$, which is known as initial phase.



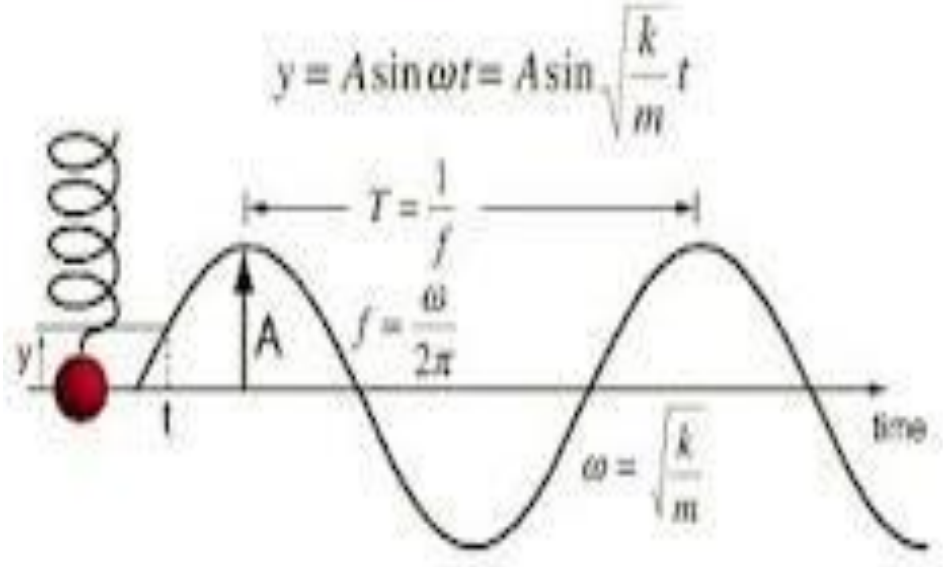



# Time period, frequency and angular frequency of a body executing SHM

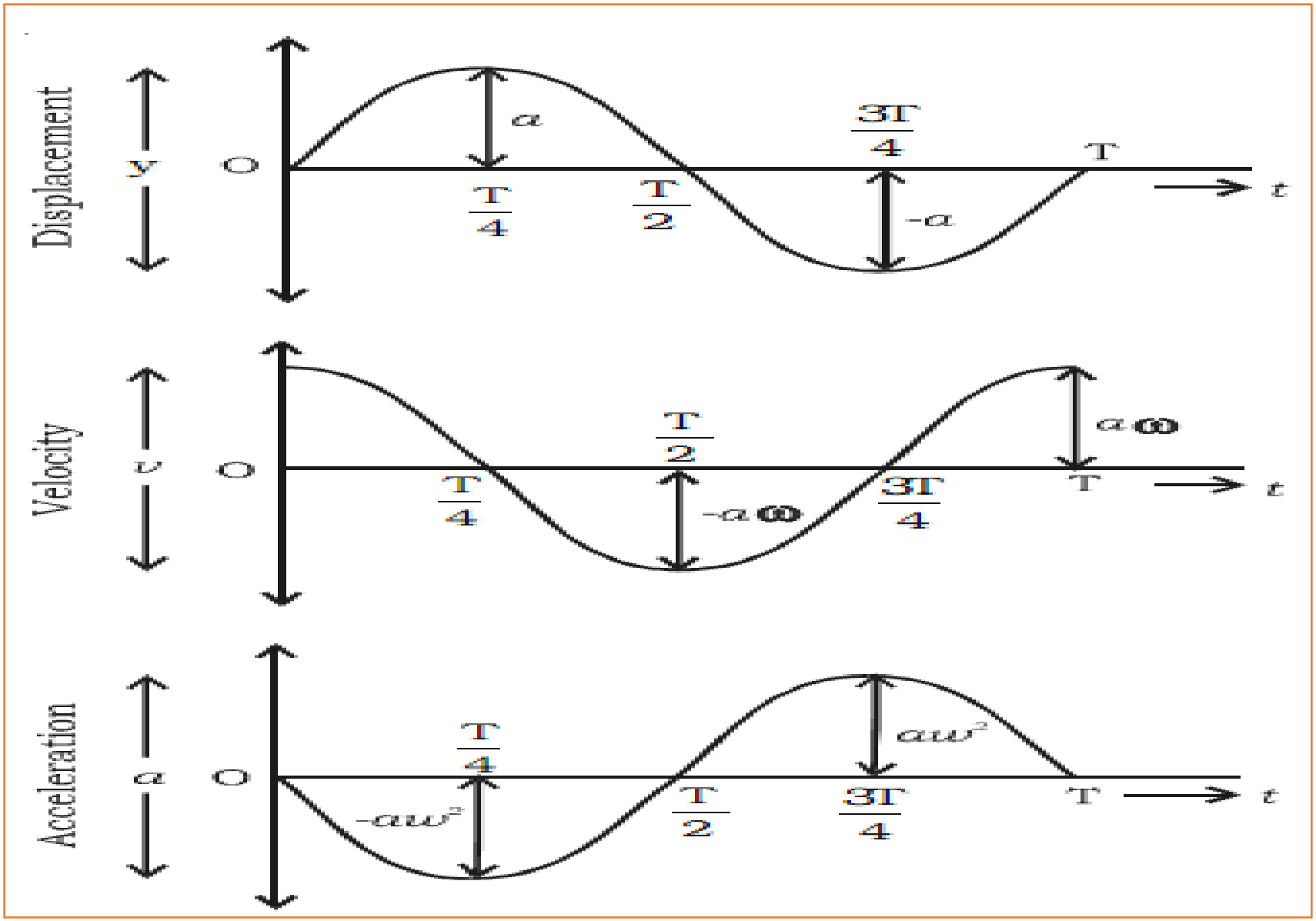
**Time period:** $$T = \frac{2\pi}{\omega} = 2\pi\sqrt{\frac{m}{k}}$$

**Frequency:** $$f = \frac{1}{T} = \frac{1}{2\pi}\sqrt{\frac{k}{m}}$$

**Angular Frequency:** $$\omega = 2\pi f = \sqrt{\frac{k}{m}}$$



# Graphical representation of SHM



**Problem:** The velocity of a particle executing simple harmonic motion are 4 cm $s^{-1}$ and 3 cm $s^{-1}$, when its distance from the mean position is 2 cm and 3 cm respectively. **Calculate its amplitude, angular velocity and time period.** **Ans.: 3.9279 cm, 1.1832 cm $s^{-1}$ and 5.31 s.**

# Energy of a body executing S.H.M.

## Potential energy of the particle

$$We\,have\,the\,displacement\,equation,\; y = A\sin\left(wt+\phi\right)$$

$$P.E. = -\int_0^y F.dy = -\int_0^y (-ky).dy = k\int_0^y y\;dy = k\;\frac{y^2}{2}$$

$$P.E. = \frac{1}{2}k\;A^2\sin^2\left(wt+\phi\right)$$

Kinetic energy of the particle

$$K.E. = \frac{1}{2}mv^2$$

$$K.E. = \frac{1}{2}m\left(\frac{dy}{dt}\right)^2$$

$$K.E. = \frac{1}{2}k\;A^2\cos^2(wt+\phi)$$

# Energy of a body executing S.H.M.

$$K.E. = \frac{1}{2} k\, A^2 \cos^2(wt + \phi)$$

Total energy of the particle

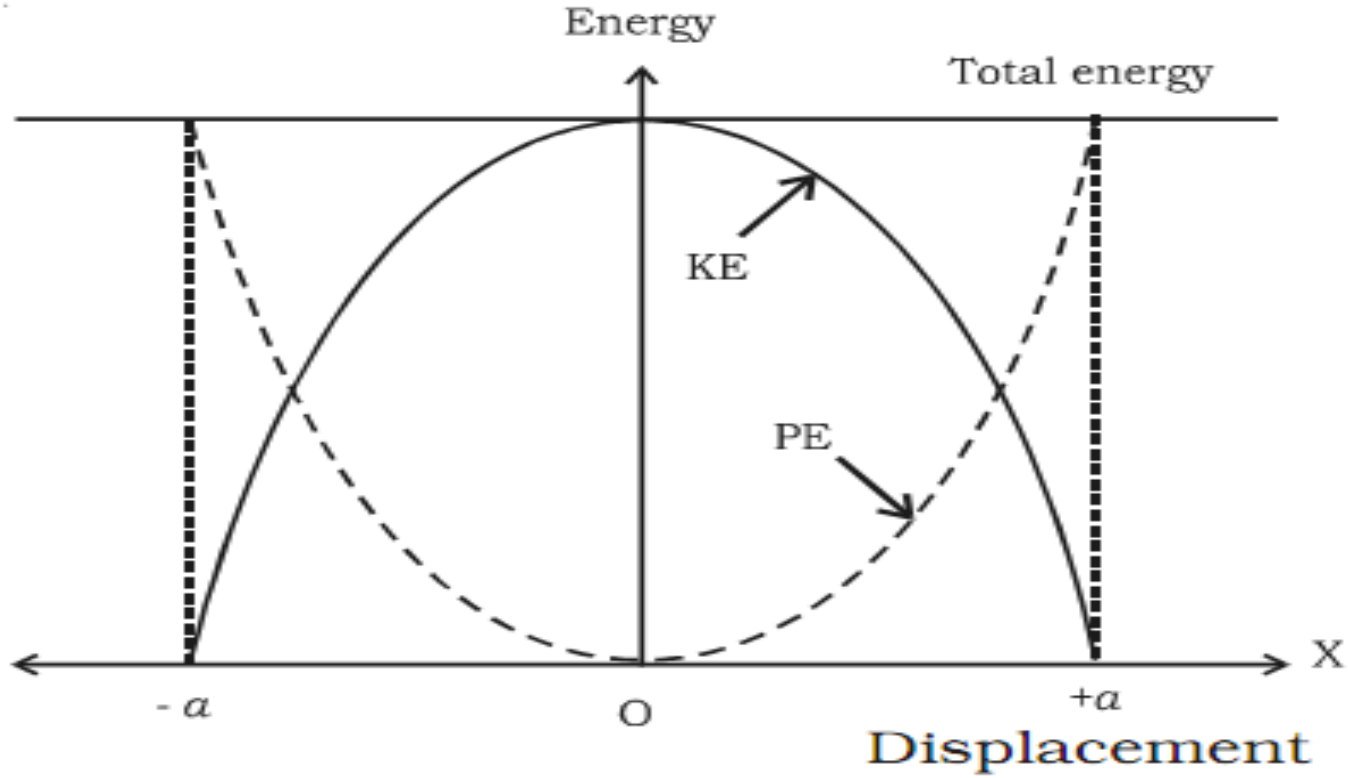
$$E = K.E. + P.E.$$

$$E = \frac{1}{2} k\, A^2$$

$$E \infty A^2$$

Total energy of a particle executing SHM is proportional to the square of the amplitude of the motion.

# *Energy – displacement graph*




| y | 0 | $\frac{a}{2}$ | $a$ | $-\frac{a}{2}$ | $-a$ |
|---|---|---|---|---|---|
| Kinetic energy | E | $\frac{3}{4}$ E | 0 | $\frac{3}{4}$ E | 0 |
| Potential energy | 0 | $\frac{1}{4}$ E | E | $\frac{1}{4}$ E | E |

A body of mass 2 kg executing SHM has a displacement $y = 3 \sin\left(100\,t + \frac{\pi}{4}\right)$ cm. Calculate the maximum kinetic energy of the body.

**Solution :** *Comparing with equation of SHM*

$$y = a \sin(\omega t + \phi_o)$$

$$a = 3 \text{ cm} = 3 \times 10^{-2} \text{ m}, \ \omega = 100 \text{ rad s}^{-1}, \ m = 2 \text{ kg}$$

$$y = 3 \sin\left(100\,t + \frac{\pi}{4}\right)$$

$$\text{Maximum kinetic energy} = \frac{1}{2} m a^2 \omega^2$$

$$= \frac{1}{2} \times 2 \times (0.03^2 \times 100^2)$$

$$\text{Maximum kinetic energy} = 9 \text{ joule}$$

**Problem:** A particle performs simple harmonic motion given by the equation $y = 20\sin(\omega t + \alpha)$. If the time period is 30 seconds and the particle has a displacement o f 10 cm at t = 0 seconds, find (i) epoch, (ii) the phase angle at t = 5 seconds .

# Thank You

Thanks for your attention